

Visionary

16+3 · BEHAVIOR protocol steal · no Discord/Drive

Steal DROID's scene-churn protocol and diversity-ablation discipline; steal π -series models — don't ship 2024 ResNet diffusion as your 2026 entry.

Spine: Same Franka · many scenes · protocol > vibes · Fig.10 before Fig.8 · 2-16-8 reactive · steal protocol into BEHAVIOR

OXE = who · DROID = where · BEHAVIOR = long-horizon household progress

Talk arc — Visionary

BEHAVIOR needs · steal · map · motion · follow-ups · narrative

01-03

Need + narrative

New houses/layouts/long horizons · OXE/DROID/BEHAVIOR

04-07

Steal cards

Protocol · Fig.10 discipline · GUI→BDDL · diversity card

08-10

Motion literacy

16/8 · race on $\pi_{0.5}/N1.7$ · Cook Lentils lesson

11-13

Follow-ups + apps

0-demo ZS · fusion · wrist · action-rep · new apps

14-16

Exam + tattoo + close

No Discord/Drive · midterm bite

BEHAVIOR needs

New houses / layouts / long horizons

New houses

Survive layout families never seen in train

New layouts

Furniture / kitchen / clutter reconfigurations

Long horizons

Stage-rich BDDL tasks, not single pick-place

Distractors

Object clutter + lighting + cam variation

Embodiment fixed

Maximize scene & viewpoint entropy under one stack

Anti-bias collect

Protocolized sampling — not easiest-task vibes

One narrative line

Who / where / long-horizon progress

OXE = who

Embodiment soup — which robot bodies exist in the prior

DROID = where

Scene soup — authentic rooms/buildings under one arm

BEHAVIOR = progress

long horizon household when both lessons meet stage structure

Steal 1 — scene-churn protocol

Anti-bias collection knobs

Time-cap churn

$\leq \sim 20$ min / ~ 100 traj per scene $\rightarrow$ forced relocation

Random task GUI

Collector-authored list $\rightarrow$ random sample (cuts easy-task bias)

View / lighting augs

Nudge base, remount cams, lighting, clutter edits

Success flags + fails

Keep fails for recovery research; exclude from success mixes

Shared stack

Identical embodiment $\rightarrow$ comparable scene entropy metrics

Diversity card

Publish layout families · distractors · cam jitter · splits

Steal 2 — Fig. 10 ablation discipline

Matched-demo many-scene vs few-scene BEFORE “more hours”

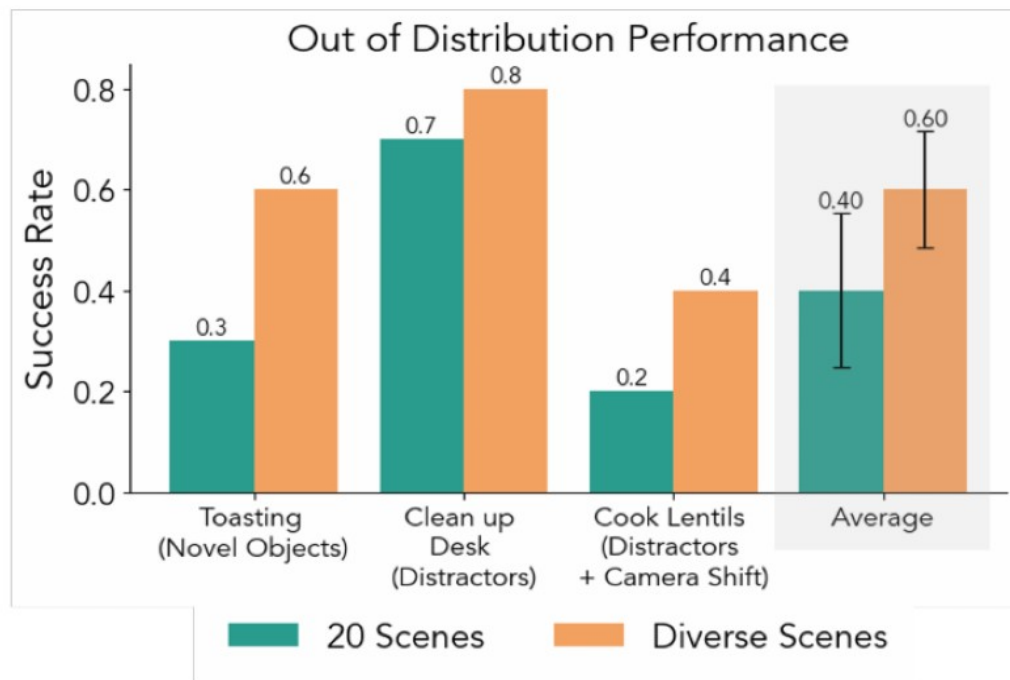


Fig. 10: **How important is the scene diversity in DROID?** We find that co-training on a subset of DROID with diverse scenes has higher OOD performance than co-training on a subset of DROID with only 20 scenes, suggesting the scene diversity of DROID is one of the

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BEHAVIOR map

- Matched demo count
- Many-scene vs few-scene arms
- Report OOD / new-house SR
- Don't claim “needed hours”
- until diversity arm loses
- Publish negative transfers
- Log scene cardinality always

Steal 3 — map GUI → BDDL ops

Concrete challenge operations

GUI random tasks



Random BDDL task sampling per scene

Scene time-cap / move robot



Forced relocation across challenge scenes

Lighting / clutter augs



Distractor + lighting jitter in sim & real

Success flagging



Stage success detectors + fail retention

Multi-instruction labels



Paraphrase / language coverage column

Fig. 10 matched-N



Challenge ablation card (many vs few layouts)

Motion literacy — 16/8 baseline

Race on $\pi_{0.5}$ / N1.7 — not 2024 ResNet DP

obs=2 / pred=16 / action=8 @ 15 Hz · commit_s $\approx$ 0.53 s

Use as literacy baseline & ablation anchor — then ship modern chunk / flow / FAST heads.

Keep

Chunking literacy ·
receding-horizon habit ·
log commit_s

Upgrade

$\pi_{0.5}$ / N1.7 / OpenVLA-FAST
as race heads

Ablate

$\text{pred} \in \{8, 16, 32\} \times \text{exec} \in \{4, 8\}$
on long BDDL

Forbid

Shipping paper ResNet DP
as 2026 BEHAVIOR entry

Cook Lentils lesson for BEHAVIOR

Stage completion under OOD distractors + camera shift

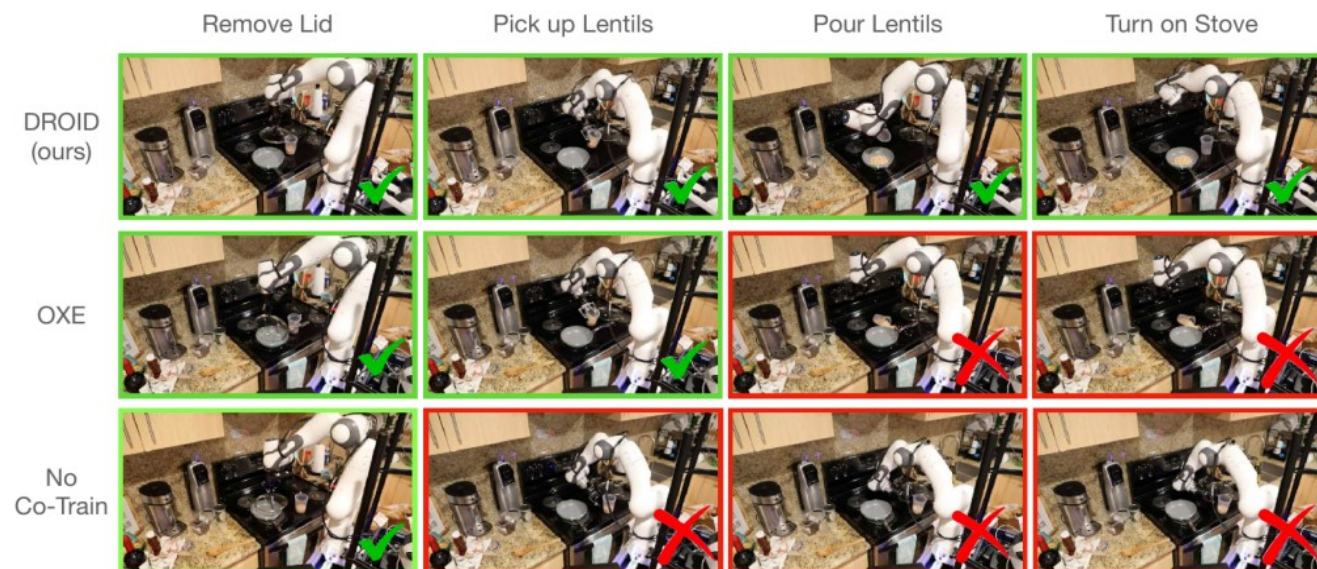


Fig. 9: Representative policy rollout examples on the most challenging “Cook Lentils” task in the kitchen of one of the authors. Qualitatively, we find that policies co-trained with DROID perform smoother, more precise motions, allowing them to solve long-horizon tasks like lentil cooking even in the presence of unseen distractor objects. In contrast, policies trained only with in-domain demonstrations or co-trained with Open-X data [39] struggle with long-horizon tasks and out-of-distribution evaluation settings. See <https://droid-dataset.github.io> for rollout videos.

Transfer

- Long BDDL = multi-stage
- Measure stages completed
- OOD = new house + distractors
- Diverse prior helps finish
- after 1-2 stages fail
- Pair with modern chunk head
- Not hierarchical-planner proof

Follow-up research agenda

Six bets after DROID

1 Zero-shot new scene

No in-domain demos — the open hole authors admit

2 Fusion done right

DROID + OXE + web/VLM — avoid naïve concat

3 When scene tail hurts

Characterize negative transfer from long tail

4 Wrist / App. G

3D-calibrated & wrist-centric policies

5 Action-rep × fit

Diffusion / FM / FAST vs naïve bins on DROID

6 BEHAVIOR diversity card

Layouts · distractors · cam jitter · instance splits

New applications

Where the prior pays rent

Household Franka assistants

Open DROID prior + hour-scale personalization

Cross-campus ZS rooms

FAST-style shared eval rooms on public Franka

3D-aware manip

Gaussian / stereo scenes from extrinsics + App. G

Data-quality tooling

Calib QA · success detectors · lang relabel

Education

Reproducible “second robot” teaching co-training

Challenge ops

Scene-rotation schedules + diversity scoreboards

Anti-patterns → fixes

Don't drink the river; don't ship 2024 DP

✗ Uniform raw dump into BEHAVIOR

✗ "More hours" without matched-N ablate

✗ Ship ResNet DP as 2026 entry

✗ Ignore action-rep × fit

✗ Skip wrist / calib you already paid for

✗ Discord/Drive checklist as plan

✓ Curate mix + diversity card + kill switch

✓ Run Fig. 10-style many vs few scenes first

✓ Race $\pi_{0.5}$ / N1.7 / FAST chunk heads

✓ Monitor per-shard action loss (OpenVLA lesson)

✓ Ablate wrist + App. G 3D priors

✓ Protocol ops + published diversity card only

Reminder — aggregate lift exists

Fig. 8 after Fig. 10 in any causality talk



Fig. 8: Does DROID Improve Policy Performance and Robustness? We find that across all our evaluation tasks, co-training with DROID significantly improves both in distribution and OOD performance over both no co-training and co-training with the Open-X dataset. We compare success rate averaged across all tasks with standard error, and find DROID outperforms the next best method by 22% absolute success rate in-distribution and by 17% out of distribution.

In Figure 10 we observe that using the split of the dataset with more diverse scenes yields better performance in the OOD evaluation setting. By comparing Figure 10's individual task performances with the corresponding tasks in Figure 8, we also see that the performance of co-training with the full DROID dataset matches or outperforms the performance with the subsampled dataset on all three tasks. These results suggest that the strength of DROID lies in its size and especially in

We hope that DROID can be a catalyst for research on general-purpose robot manipulation policies that are able to generalize to a broad range of tasks and scenes. In this work, we showed one example for leveraging DROID to boost policy performance, but there are many open questions about how to best make use of such diverse data: how should we combine DROID with existing large-scale robot datasets and how can we train policies that perform tasks in new scenes without any

Use as

- Existence proof of co-train value
- +22% ID / +17% OOD headline
- NOT causal proof alone
- Lead with Fig. 10 matched-N
- Caveat 40k≠76k
- Then pitch BEHAVIOR steal

Visionary exam bite

Protocol + models — not 2024 DP weights

“For BEHAVIOR, steal DROID’s scene-churn protocol and diversity-ablation discipline; steal π -series models — don’t ship 2024 ResNet diffusion as your 2026 entry.”

Narrative tattoo

OXE = who the robot is · DROID = where it works · BEHAVIOR’26 = long-horizon household progress when both meet stage structure.

Numbers tattoo — pin these

Shared across roles · Nov 6 ready

76k

successful traj

350 h

teleop hours

564

scenes

52

buildings

86

task verbs

50

collectors

13 / 18

labs / robots

+22% ID

Fig. 8 lift

+17% OOD

Fig. 8 lift

7362

Fig. 10 matched N

2 / 16 / 8

obs/pred/act @15Hz

~40k ≠ 76k

co-train pool

Visionary — close

Takeaways + exam bite

1. BEHAVIOR needs new houses/layouts/long horizons — maximize scene entropy under fixed embodiment.
2. Steal scene-churn + Fig. 10 discipline; map GUI→BDDL random / relocation / distractor jitter.
3. Motion: 16/8 literacy; race on $\pi_{0.5}/N1.7$ — forbid shipping 2024 ResNet DP as entry.
4. Follow-ups: 0-demo ZS, fusion, wrist/App.G, action-rep×fit, publish a BEHAVIOR diversity card.

EXAM BITE

Steal protocol + ablation discipline; steal π models — not 2024 ResNet DP.